

Research References & IoT Specifications

Smart Waste Intelligence Platform (SWIP) • Hackathon Software Demo

1. Domain Background

Modern smart waste management leverages dynamic vehicle routing (DVRP) and fill-level sensing to cut municipal collection costs by 30-45% compared to static fixed routes.

2. Sensor Telemetry Standards

- **Ultrasonic Sensor:** Fill level 0-100% (Critical threshold $\geq 90\%$).
- **Strain Gauge Load Cell:** Weight 0-60 kg (Critical threshold ≥ 45 kg).
- **Digital Thermistor:** Temperature 15-60 C (Combustion threshold ≥ 50 C).

3. Operational Benchmarks

- Automated Mean Time to Dispatch under 1 second.
- Up to 38% fuel and truck wear reduction by eliminating unnecessary pickups of half-empty containers.